Galaba Vamsi

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Galabavamsi

Summary

Motivated Mechatronics undergraduate at IIT Bhilai with a strong foundation in software development, systems programming, and robotics. Experienced in building end-to-end systems, working across IPC, web, app, embedded, and simulation layers. Passionate about solving real-world problems through scalable software and engineering innovation.

Education

IIT Bhilai
Bachelor of Technology in Mechatronics *2023 – 2027*
Relevant Coursework: Data Structures and Algorithms, Machine Learning, Robotics Systems, C++, Python Programming, ROS2, Linear Algebra, Probability and Statistics.

Air Force School Hebbal
CBSE Class 12
Scored 96%, with a strong focus on Mathematics and Computer Science.

Air Force School Yelahanka
CBSE Class 10
Scored 94.4% with distinction in STEM subjects.

Technical Skills

Languages: Python, C, C++, JavaScript, Dart, Go

Web Development: React.js, Node.js, Three.js, WebGL, HTML5, CSS3, Tailwind CSS, Flask, Bootstrap

3D & UI/UX: Three.js, GLTFLoader, OrbitControls, Shader Effects, Figma, Blender, Adobe XD

App Development: Flutter

Robotics & Systems: ROS2, NVIDIA Isaac, Drone Development, Navigation Systems, MuJoCo

Machine Learning: TensorFlow, Computer Vision, Deep Learning, Data Structures & Algorithms

Data Tools: NumPy, Pandas

CAD Design: DipTrace, SolidWorks, AutoCAD

Simulation & Modeling: MuJoCo, CST Microwave Studio, NVIDIA Isaac Sim, Unreal Engine

Tools & Platforms: Git, GitHub, Linux/Unix Environments

Projects

IIT Bhilai & IIIT Delhi
RIS Emulator – IEEE ANTS 2025 (Submitted) *2025*
Sole developer of a research-grade, open-source RIS simulator for smart wireless environments and **6G research**. Submitted as first-author paper to IEEE ANTS 2025.
Key Contributions: Developed full simulation stack, including LoS/NLoS channel modeling, dynamic mobility, and real-time BPSK-based PHY emulation with file-based IPC.
Stack: Python, multithreading, network simulation, physical layer abstraction, signal profiling.

galabavamsi.github.io/portfolio
3D Interactive Portfolio Website *2025*

Designed and developed a fully interactive 3D portfolio using **Three.js**, **GLTFLoader**, and **OrbitControls**, showcasing engineering projects in a cyberpunk-themed interface. Features real-time animations, robotic arm simulation, CRT visual effects, and custom shaders.
Stack: JavaScript, HTML5, CSS3, WebGL, Three.js, Figma.

IIT Bhilai
Shared Action Autonomous Smart Wheelchair *2025 – Present*
Led development of a smart wheelchair integrating EEG/EMG, joystick, and voice controls for enhanced accessibility. Applied deep learning techniques for signal processing and control integration. Utilized NVIDIA Isaac Sim and MuJoCo for simulation and testing.

Serenity Mental Health App *2024*
Developing a Flutter-based mental wellness app featuring stress analysis, CV-based mood tracking, gamified journaling, fitness monitoring, and guided meditation.
Key Feature: Integrated computer vision model for real-time emotion detection from facial cues, enabling passive mood logging and personalized wellness recommendations.

IIT Bhilai
Polaris Orientation Website *2024*
Built the official freshers' orientation platform using React.js and Tailwind CSS. Featured interactive dashboards, live event updates, and user resources.

IIT Bhilai - Epsilon Club
Mars Rover Project *2024*
Worked on CAD modeling and navigation system development. Integrated ROS2 with NVIDIA Isaac Sim and MuJoCo for testing.

Experience

IIT Bhilai
Research Intern *May 2025 – Aug 2025*
Researching RIS technologies; developed simulation GUIs, assisted in hardware integration, and contributed to research publications.

IIT Bhilai
Research Intern *Jun 2025 – Jul 2025*
Designed a custom two-layer PCB for a Reconfigurable Intelligent Surface (RIS) testbed using **DipTrace** for schematic capture and layout.
Used **CST Microwave Studio** for electromagnetic simulation and analysis, validating impedance characteristics and phase behavior.
Collaborated with the research team to support RIS hardware testing and integrate simulated results into software-based emulation.

IIT Bhilai
Polaris Website Developer *May 2024 – Jul 2024*
Collaborated with faculty and students to design and deploy the official orientation website with real-time updates and intuitive UI.

Swiggy Campus CEO *Apr 2024 – Sep 2024*
Planned and executed targeted campus campaigns, led promotional strategies, and increased Swiggy visibility and adoption.

IIT Bhilai
Secretary, Film and Photography Society *Apr 2024 – Apr 2025*
Led club initiatives that improved FPS Inter-IIT ranking from 17th to 6th. Organized workshops, competitions, and nationwide participation.

IIT Bhilai
Student Mentor *May 2024 – Apr 2025*
Guided first-year students academically and socially. Promoted inclusive learning and helped students navigate college life.

Achievements

Research Publication: First-author paper on RIS Simulator submitted to *IEEE ANTS 2025*, IIIT Delhi. Developed open-source channel emulator enabling scalable 6G simulation.

PCB Design: Designed and simulated a custom two-layer RIS PCB using DipTrace and CST Microwave Studio,

supporting hardware validation for 6G testbeds.

Hackathons: Toyota Hackathon – Top 10 finalist for automated car safety system. ATL Hackathon – 2nd place for BMTC Ticketing App.

Leadership: Improved IIT Bhilai's FPS Inter-IIT ranking from 17th to 6th place in one year.

Epsilon Club: Contributed to Mars Rover Project's CAD design and system integration.

Graphic Design: Shortlisted for regional round, National Skill India Competition.

Sports: 3rd Prize – Under-19 Badminton Tournament by Decathlon and Indian Air Force.

Extracurricular Activities

Robotics: Active in Epsilon Club; contributed to Mars Rover and warehouse drone projects.

Creative Tech: Designed cyberpunk-style 3D portfolio showcasing engineering projects using WebGL and interactive animation.

Media Production: Directed and edited short films; managed social media for FPS.

Reading: Science fiction enthusiast: The Martian, Three-Body Problem, Dune.

Gaming & Design: Game design, 3D modeling using Blender and Unreal Engine; card game strategy.